VARSHITHA REDDY KANCHARLA

+91-7671968289 ◇ varshithareddy.per@gmail.com
linkedin.com/in/varshitha-reddy-838743285 ◇ github.com/varshreddyy

PROFESSIONAL SUMMARY

AI/ML engineer and 2026 B.Tech (Artificial Intelligence & Machine Learning) graduate seeking entry-level Machine Learning Engineer or Backend Engineer roles. Builds end-to-end Python systems for Natural Language Processing (NLP) and Legal AI, including LawGPT, a Retrieval-Augmented Generation (RAG) legal assistant over Indian court judgments with Legal-BERT retrieval, a local Large Language Model (LLM), and an outcome classifier trained on 10,200 labelled cases. Open-source contributor to ODS, a 7.7k-star local AI server. Five-month remote AI internship, mentored by Volkswagen engineers, building a PyTorch sequence-to-sequence chatbot and a 57K-song recommender on Django REST Framework.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning, Anurag University, Hyderabad — Sep 2022 – Jun 2026
CGPA: 7.8

Class XII (Senior Secondary), Sri Chaitanya Junior College, Hyderabad — 94% — 2022

Class X (CBSE), Sri Chaitanya High School, Hyderabad — 94% — 2020

EXPERIENCE

AI Developer Intern, Nivuna Labs, Remote — May 2025 – Sep 2025

Mentored by engineers from Volkswagen

- Built a sequence-to-sequence (seq2seq) chatbot in PyTorch with a gated recurrent unit (GRU) encoder and Luong attention decoder, trained for 50K iterations on the Cornell Movie-Dialogs corpus; served via a FastAPI endpoint with a React chat UI.
- Built a lyrics-based music recommender over 57K songs (643 artists) using term frequency–inverse document frequency (TF-IDF) and cosine similarity, exposed through a Django REST Framework API and deployed on Render — GitHub

OPEN SOURCE

ODS – Cross-Encoder Reranker Service (Pull Request #1794, under review) — Pull request

- Built a reranking service for ODS, an open-source local AI server (7.7k GitHub stars): FastAPI /rerank and /health endpoints running a configurable cross-encoder (default BAAI/bge-reranker-base), packaged as a Docker extension.
- Wrote 6 pytest unit tests with a mocked model so tests run without downloading it, and added a GitHub Actions continuous integration (CI) workflow that runs them on every pull request touching the service.
- Resolved all 4 maintainer review points: aligned Docker and runtime model defaults, made model loading injectable for tests, wired tests into CI, and narrowed error handling so API errors no longer expose internals.

PROJECTS

LawGPT – Legal AI Assistant (RAG) GitHub

- Built a RAG legal research assistant over Indian Supreme Court and High Court judgments: Legal-BERT embeddings in ChromaDB, retrieval with duplicate cases removed, and answers grounded in the retrieved text and streamed from a local LLM via Ollama (phi3).
- Trained a Logistic Regression outcome classifier on 10,200 labelled judgments (Dismissed / Allowed / Partly Allowed) using Legal-BERT embeddings; shows each prediction with its confidence score alongside the most similar precedents.
- Delivered as a three-module Streamlit app: legal research, precedent search, and case assessment.

Blood Group Classification from Fingerprints (CNN + GNN) GitHub

- Built a hybrid convolutional neural network (CNN) and graph neural network (GNN) in PyTorch and PyTorch Geometric: EfficientNet-B3 feature maps converted to a 49-node k-nearest-neighbour (k-NN) graph and classified by a Graph Attention Network (GAT) into 8 blood groups.
- Reached 79.3% test accuracy on 1,600 held-out images (8 balanced classes) using 7-view test-time augmentation; served predictions through a Flask app with image upload and webcam capture.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript
- **Machine Learning & Deep Learning:** PyTorch, PyTorch Geometric, scikit-learn, Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), Sequence-to-Sequence Models
- **Natural Language Processing (NLP) & LLMs:** Retrieval-Augmented Generation (RAG), Sentence-Transformers, Legal-BERT Embeddings, Cross-Encoder Reranking, ChromaDB, Ollama
- **Backend Development:** FastAPI, Django REST Framework, Flask, REST APIs
- **Frontend Development:** React, Vite, Streamlit
- **Databases & DevOps:** SQLite, Docker, pytest, GitHub Actions, Git, Render
- **Computer Science Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP)

CERTIFICATIONS

- ServiceNow Certified System Administrator (CSA) — Feb 2026
- ServiceNow Certified Application Developer (CAD) — Feb 2026

ACHIEVEMENTS & ACTIVITIES

- Built and pitched a full-stack web application at Visionova, a national-level hackathon.
- Led marketing and coordination for Daksha Tech Fest.